

# Low-Order Optimality Certificates and Sharp Value-Estimation Limits in Structured Quantum Compilation

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## Abstract

A compact statistic can determine whether optimization is necessary while remaining provably inadequate for the value and structure of an optimum. We establish this hierarchy for a Pauli partition compiler with a fixed structural SELECT-PREPARE-width objective. For every number of terms  $m \geq 5$ , the unary compiler is globally optimal if and only if two clause families hold: every pair gain is nonpositive, and the sum of any two disjoint pair gains plus one is nonpositive. Thus a search over all set partitions has a decision certificate whose largest clause touches four term indices. The threshold is sharp at  $m = 4$ .

Decision sufficiency does not imply value or witness sufficiency. For every  $t \geq 1$ , we construct two  $5t$ -term instances with identical ordered weights and identical complete labeled pair-gain matrices. Both strictly improve on the unary baseline, but their exact improvements are  $12t - 2$  and  $10t - 1$ . Every estimator restricted to that pair representation therefore has worst-case real additive error at least  $(2t - 1)/2$ , worst-case integer error at least  $t$ , and symmetric multiplicative factor at least  $\sqrt{(12t - 2)/(10t - 1)}$ . No uniform factor below  $\sqrt{6/5}$  is possible. The same fiber forces incompatible optimizer structures: one family requires a triple block, whereas the other uses only pairs and singletons.

The limitation persists at high order. For every  $m \geq 5$  and  $L \geq 1$ , two instances agree on every labeled common-factor count through order  $m - 2$  but have improvements separated by  $[m(\lceil \log_2 m \rceil + 1) - 1]L$ . Möbius inversion shows that every nonzero integer trade preserving all proper labeled marginals is an integer multiple of the parity trade and touches all  $2^{m-1}$  Boolean cells. The result separates exact low-order decision from value estimation and optimizer recovery without invoking computational hardness.

**Keywords:** quantum compilation; low-order certificates; information lower bounds; minimax estimation; Markov bases; Möbius inversion

## 1. Introduction

Global combinatorial optimization supports several distinct downstream questions: whether a baseline is optimal, how much improvement is available, what structure an optimizer must contain, and whether a feature representation approximates the optimum uniformly. A representation can be complete for one question and incomplete for another.

We give an exact scalable example in structured quantum compilation. The compiler partitions  $m$  Pauli terms into arbitrary blocks, extracts common Pauli factors, and pays a structural cost. Although its candidate space contains every set partition, unary optimality is decided by inequalities involving at most four indices. This makes pair information appear unusually powerful. Our remaining theorems locate its exact limits.

The contributions are:

1. a four-index decision theorem for every  $m \geq 5$ , with an exact four-term counterexample;
2. two pair-indistinguishable product families with unbounded additive value gap and forced triple-versus-pair optimizer structure;
3. exact real, integer, symmetric multiplicative, and one-sided minimax lower bounds;
4. a construction showing that interactions through order  $m - 2$  do not determine exact value; and
5. a primitive-kernel theorem showing that the corresponding invisible difference trade must be the dense parity direction.

## 2. Compiler and objective

An instance is an ordered tuple of nonidentity Pauli strings  $p_1, \dots, p_m$ . Let  $w_i$  be the weight of term  $i$ , let  $W = \sum_i w_i$ , and let  $f(S)$  count the columns on which every term in a nonempty block  $S$  has the same nonidentity Pauli.

The compiler chooses a set partition  $\Pi$  and factor/ancilla options. Under the fixed equal-weight structural objective, factoring and shared width dominate their alternatives. For  $|\Pi| \geq 2$ , the reduced cost is

$$\begin{aligned} C(\Pi) = 2m + |\Pi| - 3 + \sum_{S \in \Pi} d(|S|) + \max_{S \in \Pi} b(|S|) \\ + \sum_{S \in \Pi} [2f(S) + (b(|S|) + 2)(w(S) - |S|f(S))], \end{aligned}$$

where  $b(s) = \lceil \log_2 s \rceil$ ,  $b(1) = 0$ , and

$$d(1) = 0, \quad d(s) = d(\lceil s/2 \rceil) + d(\lfloor s/2 \rfloor) + s - 2.$$

The unary incumbent has cost  $C_U = 2W + 3m - 3$ . The one-block flag convention is evaluated separately. These are structural compiler costs, not physical T counts, depth, runtime, qubits, or fault-tolerant overhead.

## 3. A four-index decision certificate

For every pair, define

$$g_{ij} = 4f(\{i, j\}) - (w_i + w_j).$$

Let  $\mathcal{P}_4(m)$  require

$$g_{ij} \leq 0$$

for every pair and

$$g_{ij} + g_{k\ell} + 1 \leq 0$$

for every two disjoint pairs.

**Theorem 1.** For every  $m \geq 5$ ,

$$\min_{\Pi} C(\Pi) = C_U \iff \mathcal{P}_4(m).$$

**Proof outline.** Expanding  $C_U - C(\Pi)$  gives an exact sum of block gains. For a block of size at least three, a matching or cycle decomposition bounds its gain by sums of pair gains. The first clause controls isolated pair gains. The second controls the only way two disjoint pair blocks can jointly evade the single-pair bound; integrality supplies the extra unit. The exceptional one-block convention is bounded separately when  $m \geq 5$ . Conversely, every failed clause directly constructs either a profitable pair partition or a profitable two-pair partition.  $\square$

The four-term instance

$$XXII, \quad XYII, \quad XZII, \quad XIXX$$

satisfies both clause families, yet has  $C_U = 27$  and one-block cost 23. The term-count threshold is therefore sharp.

## 4. Complete pair information, different values and optimizers

For every  $t \geq 1$ , construct  $t$  disjoint five-term gadgets. Families  $A_t$  and  $B_t$  have the same ordered term weights and the same common-factor count for every labeled pair, hence the same complete pair-gain matrix. Both violate  $\mathcal{P}_4$  and strictly beat the unary baseline. Exact decomposition yields

$$\Delta_A(t) = 12t - 2, \quad \Delta_B(t) = 10t - 1.$$

Their value gap is  $2t - 1$ . Every optimum in  $A_t$  contains a distinguished triple block and one pair per gadget. Every optimum in  $B_t$  uses only pairs and singletons. Thus complete pair information determines neither the exact improvement nor the presence of a triple block in an optimizer.

## 5. Exact minimax consequences

Let  $\Phi$  be any deterministic real-valued estimator whose input is exactly the term count, ordered weights, and complete labeled pair-gain matrix. The two fiber members have identical inputs, so  $\Phi$  returns one value  $y_t$  for both.

**Theorem 2 (additive minimax radius).** For every  $t \geq 1$ ,

$$\max\{|y_t - \Delta_A|, |y_t - \Delta_B|\} \geq \frac{2t-1}{2}.$$

The midpoint attains equality. If the estimator must return an integer, the minimum worst-case error is exactly  $t$ .

**Proof.** Two real values at distance  $2t-1$  cannot both lie within a smaller radius of one estimate. The integer radius is the ceiling of half the distance.  $\square$

## 5.1 Symmetric multiplicative estimation

For positive improvements, use the convention

$$\Delta/\rho \leq y \leq \rho\Delta, \quad \rho \geq 1.$$

**Theorem 3.** A common estimate valid for both fiber members requires

$$\rho \geq \sqrt{\frac{\Delta_A}{\Delta_B}} = \sqrt{\frac{12t-2}{10t-1}}.$$

The geometric mean attains equality. The bound increases with  $t$  and tends to  $\sqrt{6/5}$ . Hence no estimator using only the stated pair information has a uniform symmetric factor strictly below  $\sqrt{6/5}$  on this family.

For one-sided certificates, a common upper estimate must be at least  $\Delta_A$  but at most  $\alpha\Delta_B$  to approximate  $B_t$ ; the same ratio applies to a common lower estimate. Thus the asymptotic one-sided factor is at least  $6/5$ . These are information lower bounds, not computational hardness results.

## 6. Proper interactions still miss exact value

Fix  $m \geq 5$ , set  $q = m-1$ , and distinguish one anchor from  $q$  variable terms. Use every anchor-plus-variable support from one parity class in one instance and the opposite parity class in the other, each with multiplicity  $L \geq 1$ . Add identical all-term padding that makes the single block uniquely optimal in both instances.

The resulting pair agrees on ordered weights and on every labeled common-factor count for subsets of size at most  $m-2$ , yet

$$\Delta(A) - \Delta(B) = [m(\lceil \log_2 m \rceil + 1) - 1] L.$$

For fixed  $m$ , the ambiguity is unbounded in  $L$ .

## 7. The unique invisible direction

Represent a trade column by a subset of  $[q]$ . Let  $\delta : 2^{[q]} \rightarrow \mathbb{Z}$  be the signed multiplicity difference, and define its upper marginal by

$$M(T) = \sum_{S \supseteq T} \delta(S).$$

Equality of all proper labeled marginals is  $M(T) = 0$  for every proper  $T \subsetneq [q]$ .

**Theorem 4 (proper-marginal kernel).** If every proper upper marginal vanishes, then

$$\delta(S) = (-1)^{q-|S|}c, \quad c = \delta([q]).$$

**Proof.** Möbius inversion on the Boolean lattice expresses  $\delta(S)$  as the alternating sum of upper marginals over supersets of  $S$ . Only the top marginal can remain.  $\square$

If  $c \neq 0$ , every Boolean cell is used and exactly half occur on each signed side. A primitive integer trade therefore has mass at least  $2^{q-1} = 2^{m-2}$  on each side. The parity construction with  $L = 1$  attains this bound. The statement proves minimality of the *difference trade*; it does not prove that the common padding is minimal.

## 8. Relation to prior work

Markov-basis and hierarchical-model theory [1-5] owns the generic language of fibers, marginal-preserving moves, toric ideals, and higher-order interactions invisible to lower marginals. Boolean-lattice Möbius inversion is classical.

The residual result is the exact conjunction for the fixed compiler objective: a four-index certificate decides global unary optimality for all  $m \geq 5$ , the same pair representation has exact unbounded and multiplicative value limits, optimizer block structure is not identifiable, and even order- $m - 2$  information misses exact value through the primitive parity direction.

## 9. Reproducibility and limitations

The four-term witness, pair-indistinguishable gadgets, product formulas, minimax optima, and Boolean-lattice kernel have been checked with independent exact implementations; the analytic arguments provide all-parameter authority.

The results apply to the stated structural objective. The decision theorem does not reconstruct an optimizer. The minimax bounds apply to estimators restricted to the exact declared representation. The high-order construction uses conservative common padding whose minimality is open. The work proves information nonidentifiability, not a complexity-class lower bound. Transfer to other compiler grammars or physical resource models remains open.

## 10. Conclusion

Information sufficiency is query-dependent. Complete pair information decides whether the unary compiler is optimal, yet it cannot recover the improvement value within arbitrarily small additive or fixed multiplicative error and cannot determine optimizer block structure. Even all labeled interactions below the top two orders fail to determine exact value, and the invisible integer direction is necessarily dense.

A perfect decision statistic is therefore not automatically a useful value statistic. In this compiler, the distinction is exact, scalable, and independent of computational assumptions.

## Data and code availability

Exact implementations reproducing the four-term witness, paired gadget families, product formulas, minimax calculations, and Boolean-lattice kernel accompany the submission source. They are verification aids; the displayed arguments carry the all-parameter claims. A permanent archival identifier must be inserted before final upload.

## References

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